

Year 7 • Chapter 7 • Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 0.7
- Ex 1b: 34.89
- Ex 2a: 8 hundredths
- Ex 2b: 3.998, 34.89, 34.98, 36.86, 38.141, 340
- Ex 3a: 1.293, 1.239, 1.1015, 1.015, 1.006
- Ex 3b: e.g. 2.355

Easy (drill)

- | | | |
|----------|-------------------|-------------------|
| 1. 8.42 | 12. 3.5 | 23. 9 hundredths |
| 2. 1.69 | 13. 5.6 | 24. 5 ones |
| 3. 2.7 | 14. 4.71 | 25. 5 hundredths |
| 4. 8.8 | 15. 9.15 | 26. 1 thousandths |
| 5. 3.92 | 16. 3.2 | 27. 8 tenths |
| 6. 9.64 | 17. 2 thousandths | 28. 3 thousandths |
| 7. 3.39 | 18. 7 tenths | 29. 7 ones |
| 8. 8.22 | 19. 3 tenths | 30. 2 thousandths |
| 9. 6.91 | 20. 9 thousandths | 31. 4 thousandths |
| 10. 8.55 | 21. 8 thousandths | 32. 2 thousandths |
| 11. 1.9 | 22. 8 ones | |

Medium (drill)

- | | | |
|------------------------------|------------------------------|-------|
| 1. 1.34, 1.4, 1.579,
1.85 | 5. 1.68, 1.75,
1.824, 1.9 | 9. > |
| 2. 2.15, 2.685, 2.8,
2.89 | 6. 2.5, 2.53, 2.672,
2.98 | 10. < |
| 3. 3.64, 3.72, 3.8,
3.965 | 7. 3.51, 3.578, 3.81,
3.9 | 11. > |
| 4. 5.1, 5.38, 5.52,
5.96 | 8. 0.8, 0.897, 0.93,
0.96 | 12. < |
| | | 13. > |
| | | 14. > |
| | | 15. > |
| | | 16. < |

Challenging

- 1.293, 1.239, 1.1015, 1.015, 1.006
- e.g. 2.355
- 12.018 s

Extension

- 0.605
- 9 (4.31 to 4.39)

Exit ticket

- Level 1: 0.4
- Level 2: 5.02, 5.2, 5.22
- Level 3: 0.33, 0.303, 0.3, 0.033

7.02 Decimals and fractions (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{2}$
- Ex 1b: $\frac{89}{100}$
- Ex 2a: $\frac{111}{250}$
- Ex 2b: 0.4
- Ex 3a: 0.07
- Ex 3b: 0.011

Easy (drill)

- | | | |
|---------------------|---------------------|-----------|
| 1. $\frac{1}{25}$ | 10. $\frac{1}{100}$ | 20. 0.1 |
| 2. $\frac{1}{50}$ | 11. $\frac{3}{10}$ | 21. 0.86 |
| 3. $\frac{33}{100}$ | 12. $\frac{17}{25}$ | 22. 0.4 |
| 4. $\frac{43}{100}$ | 13. $\frac{1}{5}$ | 23. 0.24 |
| 5. $\frac{9}{100}$ | 14. $\frac{11}{25}$ | 24. 0.89 |
| 6. $\frac{9}{10}$ | 15. $\frac{3}{50}$ | 25. 0.56 |
| 7. $\frac{4}{5}$ | 16. $\frac{4}{25}$ | 26. 0.15 |
| 8. $\frac{7}{100}$ | 17. 0.47 | 27. 0.98 |
| 9. $\frac{41}{50}$ | 18. 0.35 | 28. 0.188 |
| | 19. 0.96 | 29. 0.017 |
| | | 30. 0.404 |
| | | 31. 0.92 |
| | | 32. 0.38 |

Medium (drill)

- | | | |
|-----------------------|------------------------|------------------------|
| 1. $\frac{177}{250}$ | 9. $\frac{171}{200}$ | 17. 3 $\frac{37}{100}$ |
| 2. $\frac{927}{1000}$ | 10. $\frac{619}{1000}$ | 18. 2 $\frac{3}{4}$ |
| 3. $\frac{47}{125}$ | 11. $\frac{963}{1000}$ | 19. 3 $\frac{11}{20}$ |
| 4. $\frac{111}{200}$ | 12. $\frac{951}{1000}$ | 20. 4 $\frac{1}{4}$ |
| 5. $\frac{661}{1000}$ | 13. $\frac{157}{1000}$ | 21. 3 $\frac{33}{50}$ |
| 6. $\frac{143}{500}$ | 14. $\frac{277}{1000}$ | 22. 2 $\frac{41}{50}$ |
| 7. $\frac{487}{1000}$ | 15. $\frac{413}{500}$ | |
| 8. $\frac{409}{500}$ | 16. $\frac{759}{1000}$ | |

Challenging

- 0.6
- 0.35
- 0.036

Extension

- 0.075
- 4 $\frac{3}{40}$

Exit ticket

- Level 1: $\frac{7}{10}$
- Level 2: 0.23
- Level 3: $\frac{7}{20}$

7.03 Adding and subtracting decimals

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 16.88
- Ex 1b: 2.379
- Ex 2a: 27.595
- Ex 2b: 104.49
- Ex 3a: 396.51
- Ex 3b: 65.355

Easy (drill)

- | | | |
|---------|----------|-----------|
| 1. 12.5 | 9. 9.3 | 17. 12.88 |
| 2. 3.4 | 10. 7.8 | 18. 8.19 |
| 3. 8 | 11. 2.2 | 19. 13.97 |
| 4. 12.2 | 12. 12.9 | 20. 1.73 |
| 5. 10.3 | 13. 10.4 | 21. 2.21 |
| 6. 0.8 | 14. 14.9 | 22. 1.91 |
| 7. 12.6 | 15. 12.4 | 23. 7.29 |
| 8. 15.9 | 16. 1.4 | 24. 9.6 |

25. 6.86	28. 1.08	31. 12.04
26. 5.53	29. 9.94	32. 6.92
27. 4.82	30. 15.48	

Medium (drill)

1. 35.32	9. 129.56	17. 4.16
2. 179.46	10. 40.01	18. 3.97
3. 34.99	11. 109.41	19. 10.99
4. 19.3	12. 369.41	20. 13.4
5. 17.4	13. 321.99	21. 3.16
6. 41.11	14. 10.16	22. 8.41
7. 18.02	15. 5.56	
8. 38.22	16. 150.1	

Challenging

- 65.355
- 396.51
- \$29.75

Extension

- 8.43
- $6.42 + 5.31 = 11.73$ (or similar)

Exit ticket

- Level 1: 4.3
- Level 2: 6.55
- Level 3: 19.08

7.04 Multiplying and dividing by powers of 10 (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: 75.4
- Ex 1b: 754
- Ex 2a: 7 540
- Ex 2b: 1.39
- Ex 3a: 0.139
- Ex 3b: 0.0139

Easy (drill)

1. 4 120	12. 9.08	23. 17.47
2. 595	13. 4.48	24. 81.92
3. 66	14. 18.3	25. 73.67
4. 6.55	15. 295	26. 734.1
5. 9 100	16. 76 400	27. 33.32
6. 92 300	17. 11.37	28. 8.182
7. 831	18. 3.605	29. 737.8
8. 932	19. 94.14	30. 1.841
9. 49 300	20. 6.96	31. 159
10. 43 100	21. 7.035	32. 0.5658
11. 2 490	22. 956.4	

Medium (drill)

1. 0.00635	9. 71 700	17. 10
2. 43	10. 8.9	18. 10
3. 0.00705	11. 63	19. 1 000
4. 963	12. 37.9	20. 100
5. 99.6	13. 0.686	21. 10
6. 9 600	14. 0.0518	22. 10
7. 27.5	15. 0.00123	23. 10
8. 0.0014	16. 5 410	24. 1 000

Challenging

- \$135
- \$8.75
- 0.096 mm

Extension

- 7 200

2. 0.16

Exit ticket

- Level 1: 38
- Level 2: 0.526
- Level 3: 49

7.05 Using a known product (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: 319.2
- Ex 1b: 31.92
- Ex 2a: 319.2
- Ex 2b: 3.192
- Ex 3a: 319.2
- Ex 3b: 0.3192

Easy (drill)

1. 0.972	9. 0.056	17. 0.048
2. 0.1178	10. 0.028	18. 0.014
3. 10.12	11. 0.033	19. 0.0072
4. 8.12	12. 0.18	20. 0.9
5. 1.189	13. 0.008	21. 0.27
6. 64.4	14. 0.1	22. 0.6
7. 0.594	15. 0.081	23. 3.6
8. 0.0342	16. 1.8	24. 0.0008

Medium (drill)

1. 1.8	5. 2.2	9. 4 420
2. 0.24	6. 1.9	10. 12 250
3. 3.4	7. 4.7	11. 660
4. 3.5	8. 0.45	12. 682

Challenging

- 45
- 45
- 0.045

Extension

- 17
- 24

Exit ticket

- Level 1: 32.2
- Level 2: 3.22
- Level 3: 0.0322

7.06 Multiplying decimals (booklet p. 23)

WE DO examples (teacher)

- Ex 1a: 16.5
- Ex 1b: 0.6
- Ex 2a: 7.32
- Ex 2b: 1.3325
- Ex 3a: 0.001
- Ex 3b: 5.137

Easy (drill)

1. 26	12. 6	23. 0.08
2. 58.1	13. 40.5	24. 0.015
3. 10.8	14. 8.2	25. 0.12
4. 5.4	15. 20.7	26. 0.1
5. 65.8	16. 48.3	27. 0.002
6. 8.4	17. 0.18	28. 0.27
7. 60	18. 0.035	29. 0.007
8. 11.2	19. 0.018	30. 0.3
9. 60.8	20. 0.01	31. 0.15
10. 7.6	21. 0.06	32. 0.009
11. 25.5	22. 0.027	

1. 0.744	8. 2.408	15. 2.013
2. 1.6	9. 0.782	16. 1.587
3. 4.897	10. 2.444	17. 16.224
4. 2.352	11. 6.132	18. 8.229
5. 2.652	12. 2.94	19. 26.465
6. 1.584	13. 1.829	
7. 4.44	14. 2.47	

1. 14.4 m^2
2. \$80.33
3. 0.008

1. 0.0081
2. e.g. 0.4×0.6

1. Level 1: 8.6
2. Level 2: 0.42
3. Level 3: 3.29

7.07 Dividing decimals by whole numbers

(booklet p. 27)

1. Ex 1a: 381.2
2. Ex 1b: 12.2
3. Ex 2a: 65.2
4. Ex 2b: 0.6875
5. Ex 3a: 0.4375
6. Ex 3b: \$8.40

1. 9	12. 5.4	23. 7.16
2. 9.6	13. 8.7	24. 8.03
3. 3.8	14. 8.9	25. 9.88
4. 8.3	15. 1.1	26. 4.43
5. 5.6	16. 8.6	27. 6.89
6. 1.5	17. 8.14	28. 9.77
7. 4.9	18. 3.53	29. 1.76
8. 6	19. 1.13	30. 2.82
9. 8.7	20. 8.54	31. 3.62
10. 4.3	21. 1.47	32. 7.23
11. 7.9	22. 6.52	

1. 1	9. 1.65	17. \$6.40
2. 0.58	10. 4.3	18. \$1.48
3. 2.4	11. 0.4875	19. \$15.06
4. 0.5	12. 0.475	20. \$19.78
5. 0.3875	13. 0.625	21. \$24.85
6. 1.05	14. 1.05	22. \$20.80
7. 1.75	15. 1.4	
8. 1.2	16. 4.7	

1. \$8.40
2. 1.125 m
3. \$1.45

1. 5.6
2. 1.25 and 0.625: doubling the divisor halves the answer

1. Level 1: 3.4
2. Level 2: 2.4

3. Level 3: 1.325

7.08 Dividing by a decimal (booklet p. 31)

1. Ex 1a: 25
2. Ex 1b: 31.8
3. Ex 2a: 230
4. Ex 2b: 25.4
5. Ex 3a: 5
6. Ex 3b: 14 cups

1. 20	12. 3	23. 4
2. 11	13. 8	24. 3
3. 24	14. 2	25. 26
4. 29	15. 15	26. 15
5. 9	16. 6	27. 19
6. 11	17. 40	28. 33
7. 8	18. 22	29. 5
8. 29	19. 21	30. 33
9. 22	20. 35	31. 39
10. 24	21. 38	32. 7
11. 5	22. 32	

1. 7.9	9. 6.7	17. 74
2. 4.4	10. 5.6	18. 79
3. 8	11. 2.5	19. 75
4. 3.4	12. 1.1	20. 5
5. 2	13. 3.9	21. 52
6. 7.8	14. 8.5	22. 76
7. 2.4	15. 4.4	
8. 6.7	16. 6.9	

1. 14 cups
2. 12.5 km/L
3. 14 m

1. 180
2. the same: both are 80

1. Level 1: 20
2. Level 2: 9
3. Level 3: 4

7.09 Fractions to decimals (booklet p. 35)

1. Ex 1a: 0.8
2. Ex 1b: 0.375
3. Ex 2a: 0.6
4. Ex 2b: 0.5
5. Ex 3a: 0.45
6. Ex 3b: 0.16

1. 0.125	10. 0.04	19. 0.083
2. 0.25	11. 0.6	20. 0.06
3. 0.5	12. 0.9	21. 0.3
4. 0.875	13. 0.4	22. 0.6
5. 0.55	14. 0.95	23. 1.6
6. 0.3	15. 0.85	24. 0.8
7. 0.2	16. 0.68	25. 0.1
8. 0.02	17. 0.83	26. 0.2
9. 0.625	18. 1.16	27. 0.16

28. 0.4	30. 0.7	32. 0.03	22. 3.31	23. 1.67	24. 2.50
29. 0.5	31. 1.3				

Medium (drill)

1. 2.4	10. 6.96	18. 0.23
2. 7.8	11. 4.4	19. $\frac{4}{5}$
3. 2.88	12. 5.5	20. 0.78
4. 7.48	13. 4.125	21. 0.84
5. 2.375	14. 4.5	22. 0.6
6. 4.05	15. 9.72	23. 0.53
7. 3.625	16. 6.2	24. 0.78
8. 1.625	17. $\frac{3}{5}$	
9. 3.5		

Challenging

- 0.6, $\frac{5}{8}$, $\frac{2}{3}$, 0.67
- 0.1, 0.2, 0.3, 0.7
- 0.142857 repeating: 6 digits

Extension

- $\frac{7}{9}$
- $\frac{2}{11}$

Exit ticket

- Level 1: 0.6
- Level 2: 0.875
- Level 3: 0.3

7.10 Rounding decimals (booklet p. 39)

WE DO examples (teacher)

- Ex 1a: 406.3
- Ex 1b: 125.72
- Ex 2a: 3.90
- Ex 2b: 2345.9
- Ex 3a: 78654.06
- Ex 3b: 678.431

Easy (drill)

1. 38.8	12. 87.8	23. 44.74
2. 12.5	13. 1.1	24. 69.06
3. 70.6	14. 38.3	25. 70.74
4. 17.8	15. 49.5	26. 87.02
5. 23.4	16. 70.1	27. 12.88
6. 31.8	17. 81.85	28. 83.20
7. 74.8	18. 18.00	29. 73.45
8. 86.7	19. 4.66	30. 17.70
9. 9.2	20. 25.33	31. 95.80
10. 26.0	21. 11.97	32. 35.49
11. 3.3	22. 90.53	

Medium (drill)

1. 25	8. 81	15. 89.829
2. 36.698	9. 39.989	16. 43.331
3. 31.536	10. 60.5	17. 6.75
4. 97	11. 28.539	18. 12.67
5. 89.87	12. 72	19. 5.83
6. 64.88	13. 20.4	20. 3.64
7. 100.0	14. 73	21. 2.58

Challenging

- 0.67
- \$15.74
- 4.65 and 4.74

Extension

- 23 000
- 0.0061

Exit ticket

- Level 1: 5.7
- Level 2: 12.35
- Level 3: 0.900

7.11 Decimals and money (booklet p. 43)

WE DO examples (teacher)

- Ex 1a: \$470.20
- Ex 1b: \$1.34
- Ex 2a: \$21.75
- Ex 2b: \$38.20
- Ex 3a: \$17.85
- Ex 3b: \$8.66

Easy (drill)

1. \$11.70	10. \$12.29	19. \$0.05
2. \$9.75	11. \$5.79	20. \$28.73
3. \$9.61	12. \$36.42	21. \$29.25
4. \$22.55	13. \$1.45	22. \$2.57
5. \$29.90	14. \$17.86	23. \$9.28
6. \$6.29	15. \$45.39	24. \$5.76
7. \$20.73	16. \$30.76	25. \$0.37
8. \$19.22	17. \$0.79	
9. \$22.08	18. \$25.36	

Medium (drill)

1. \$36.45	11. \$69.18	21. \$51.30
2. \$44.00	12. \$22.48	22. \$75.10
3. \$60.05	13. \$70.95	23. \$55.85
4. \$16.08	14. \$93.50	24. \$77.20
5. \$10.90	15. \$60.70	25. \$86.25
6. \$43.20	16. \$93.80	26. \$85.65
7. \$34.86	17. \$81.10	27. \$18.30
8. \$6.84	18. \$89.15	28. \$57.35
9. \$21.40	19. \$32.75	
10. \$77.77	20. \$9.90	

Challenging

- \$38.20
- \$44.60
- 2 L for \$4.60 (\$2.30/L vs \$2.40/L)

Extension

- one of each pack: 8 pens for \$10.70
- Yes: \$1.20 per pen

Exit ticket

- Level 1: \$14.60
- Level 2: \$5.65
- Level 3: \$12.75